NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.2e Warnings for wind-sensitive operations and structures

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-14 18:16 — 2026-10-14 19:04

Stage conclusion

Project report

F Wind-sensitive operations and structural warnings

Objective

Ensemble / multi-source forecasts provide exceedance probabilities → warning decisions for wind-sensitive operations and structures such as crane lifting, bridge closures, and construction; assess how much economic value this pipeline adds over deterministic forecasts and persistence.

Current Status

10-14 20:0x Shelved (3/3 conditional support; verdict). - Failed frozen primary criterion: for sustained wind 10 m/s, 24 h lead time, and C/L 0.05–0.5 (assumed), the four-source mean probability achieved relative economic

value V = 0.266 vs. 0.156 for the best deterministic forecast—a difference of 0.111, with a 95% CI of [−0.006, 0.179]. - The advantage of probabilistic decision rules over deterministic thresholds stems primarily from the probabilistic formulation itself: at low C/L, single-source GFS probability performs just as well as the four-source mean (0.598 vs. 0.599 at 10 m/s); across C/L 0.05–0.5, the four-source mean exceeds the best single source (Pangu) by only 0.024, and at the 12 m/s threshold it underperforms single-source GraphCast. No stable incremental gain unique to AI multi-source blending was found. - Precedents: HS2 Align (2021–23) already used global NWP + local LSTM bias correction to support viaduct lifting decisions; the Severn Bridge closes based on forecast gusts. The only aspects not found in prior work are "directly driven by AI global models" and "empirically measured C/L." - By-products: threshold and wind-speed definition table (below); finding that deterministic linear bias correction dampens strong winds and underperforms raw GFS at low C/L (a cautionary note for A1 bias-correction design). - **Restart conditions**: an operational partner provides downtime cost/loss data or specific operational requirements; or A1 adds daytime operational verification hours (e.g., 18 UTC); or Node B produces a true ensemble (used only as a re-evaluation trigger). Re-evaluation takes ~10 minutes and 0 CU using existing scripts.

Operation Type / Criteria & Sources / Current Warning Practices / What We Can Add

Operation Type	Criteria & Sources	Current Warning Practices	What We Can Add
Crane lifting	3 s gust at boom tip: tower crane design limit 20 m/s, UK practical shutdown at 16.5–17 m/s; mobile crane mean 9–11 m/s; crawler crane 12–16 m/s; personnel basket 7 m/s (CPA CIG/TCIG 2507, PDF pp. 13, 20, 23, 47, citing BS EN 13001-2 and BS 7121). Converted to 10 m mean wind, this corresponds to roughly 5–10 m/s (assumption; see verification log). Chinese national standards (GB) / OSHA clauses pending verification by reviewers E/F	Real-time on-site anemometer monitoring + manufacturer manual thresholds; HS2 Align used NWP + local LSTM correction to support lifting decisions (HS2 Learning Legacy, verified); advance planning by multiplying surface forecasts by height-dependent gust factors (CPA PDF pp. 21–22); no probabilistic exceedance warning products found (Grok; not found ≠ nonexistent)	Hindcast replay: probabilistic decision rules outperform deterministic thresholds, but the gain comes from the probabilistic formulation rather than AI multi-source blending (primary criterion failed)
Bridge traffic restrictions / closures	Severn M48: tiered by forecast gusts at 30/35/40/70 kn, full bridge closure at 40 kn (National Highways official website); Mackinac: tiered by sustained winds at 20/35/50/65 mph (official website FAQ, explicitly sustained wind rather than gust); verified	Severn already decides bridge closures based on **forecast gusts**—precedent exists for forecast-driven closures	Probabilistic formulation + cost/loss economic value evaluation; no internal hindcast replay (lacking bridge-site observations)

Operation Type	Criteria & Sources	Current Warning Practices	What We Can Add
Work at height / construction	OSHA 1926.1431(k)(8)(i): >20 mph at personnel platform requires judgment by a competent person (verified); Chinese standards (secondary sources, primary text unverified): GB 55034 "suspend work at height when workplace wind reaches Beaufort scale 5 or above"; JGJ 80 "Beaufort scale 6 or above"	Primarily on-site judgment	TBD

Boundaries

- No per-experiment cap on Colab (updated by parent project at 10-14 18:3x; previous ≤20 CU cap is void); report to Node A via `--report` first if a single experiment is projected to exceed 100 CU; must run `colab stop` immediately after use; dispatch only one executor per round. Supercomputer single-job limit of ≤50 SU remains unchanged.

Finalized Decisions

- 10-14: Selected crane lifting as the initial operation type; replaced "ensemble" with an equal-weighted average of four-source probabilities (GraphCast/Pangu/FCNv2/GFS) while explicitly noting it is a proxy; froze hindcast replay criteria (see task brief). Decision.

Trials and Results

Date	Action	One-Sentence Result
10-14	Hindcast replay: OKC 12 stations in 2023, four-source mean probability vs. deterministic vs. persistence, cost/loss relative economic value V (ideas, 0 CU)	**Failed frozen primary criterion**: for sustained wind 10 m/s at 24 h lead time across C/L 0.05–0.5, V = 0.266 vs. 0.156 for best deterministic, difference 0.111, daily-block bootstrap 95% CI [−0.006, 0.179] crosses 0 (135 events). Supporting evidence is consistently positive: point estimates for multi-source probability exceed best deterministic across all 12 threshold × lead-time combinations (+0.08–0.22), with CIs not crossing 0 for 8 m/s sustained wind and 10 m/s gust; multi-source probability beats single-source GFS probability by 0.108 [0.036, 0.226]; persistence V is negative across the board. Lead independently recomputed both primary V values with identical results. Supplemental report for 6 m/s (roughly corresponding to mobile crane shutdown, report only): 24 h V = 0.492 vs. 0.320, difference 0.171 [0.119, 0.214]; 48 h difference 0.175 [0.122, 0.214] (Lead recomputation matched; if deterministic candidates also include bias-corrected Pangu/FCNv2, the best deterministic at 6 m/s is Pangu at 0.329 with a difference of ~0.163, whereas at the 10 m/s primary criterion the best deterministic remains GFS at 0.156, unchanged)

Parameter Source Table

Parameter	Value	Source or "Assumption"	Sources Checked
Crane lifting shutdown wind speed (3 s gust at boom tip)	Tower crane 16.5–17 m/s (design 20); mobile crane 9–11; crawler crane 12–16; personnel basket 7	CPA CIG/TCIG 2507 (2025) PDF pp. 13/20/23/47	, pages (none); Grok; original CPA text
10 m mean wind → boom-tip gust factor	1.65 (50 m)–1.86 (150 m); gust → gust 1.18–1.33	CPA PDF pp. 21–22 (Liebherr table, rural roughness)	Same as above

Parameter	Value	Source or "Assumption"	Sources Checked
Hindcast threshold tiers	Sustained wind 8/10/12 m/s, gust 10/13/15 m/s	Assumption: inherited from A1 tiers, roughly corresponding to tower/mobile crane shutdown per the conversion factors above	—
Cost/loss ratio C/L	0.05–0.5 (frozen criterion); sensitivity 0.01–0.10	Assumption. Reviewer E reported a typical offshore operations value of 0.1 (Carrasco 2013, cited via SECLI-FIRM 2021 slides), link 404 unverified; E roughly estimated 0.01–0.10 from daily downtime cost / accident loss without multiplying L by accident probability, biasing it low. Reviewer F found no published figures	, Grok, reviewers E/F
Chinese domestic lifting thresholds	>9.0 m/s at highest point during erection/dismantling (GB 55034 §3.4.7); ≥12 m/s for tower crane operation (JGJ 196 §4.0.9, clause does not specify averaging period or height; reviewer A reported verifying the primary text); Beaufort scale "6 or above" for lifting (JGJ 276 §3.0.14; "10 m, 10 min mean" is extrapolated from the meteorological wind-scale definition rather than stated in the standard itself)	Consistent between reviewers E and F, but both from secondary sources (JGJ 33 §§4.1.14/4.1.15, JGJ 196 §4.0.9, GB 55034 §3.4.7), primary text unverified	E/F
Relative economic value V definition	Static cost/loss model, V={min(o,r)−(h+f)r−m}/{min(o,r)−o·r}, r=C/L, climatology=0, perfect forecast=1	Zhu et al. 2002 BAMS journal p. 75 (PDF p. 3) Eqs. 5–7, card; hindcast implementation (unavoidable loss = 0) matches this formulation, verified by Lead	Card created by litwiki
Daily event definition	Exceedance at either 00 or 12 UTC on a given day	Assumption (sensitivity analysis only)	—

Open Items and Next Steps

- Grok response received and checked against primary sources. Waiting for reviewers E/F to return: verify primary texts, populate operational thresholds and C/L values, finalize the assessment table, and

submit literature conclusions.

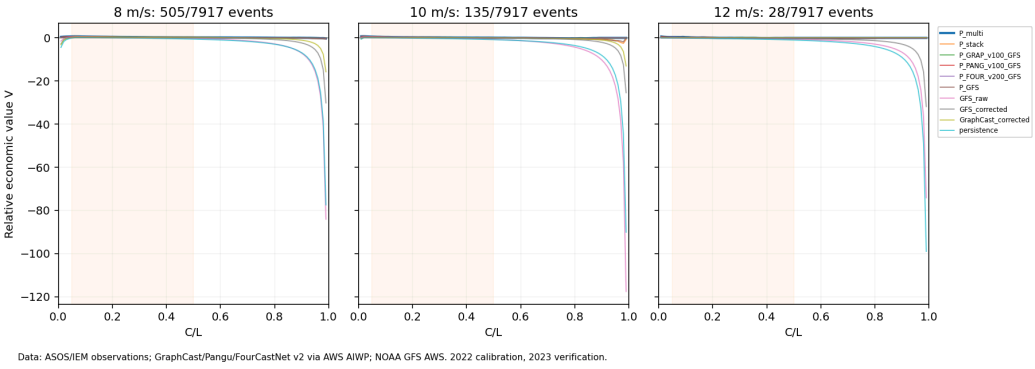
- Currently shelved, with no running tasks. Once Deep Research quota recovers, a supplemental check for precedents on "AI global models used for operational shutdown decisions" can be run (low priority).
- Pending verification (for reference only): Glasgow S2S crane-hire cost–loss study (eprints.gla.ac.uk/320158); primary texts of JGJ/GB clauses. Final "worth pursuing / shelved" determination awaits E/F clause checks so literature conclusions and hindcast interpretation can be submitted together; Lead inclination: make no claim of economic value, treat as shelved, and document restart conditions (empirical C/L, operational daytime verification hours, true ensemble).
- Limitations noted upfront: A1's 24/48 h verification times only cover 00/12 UTC (local morning/evening); 10 m wind ≠ boom-tip wind.

Literature Basis

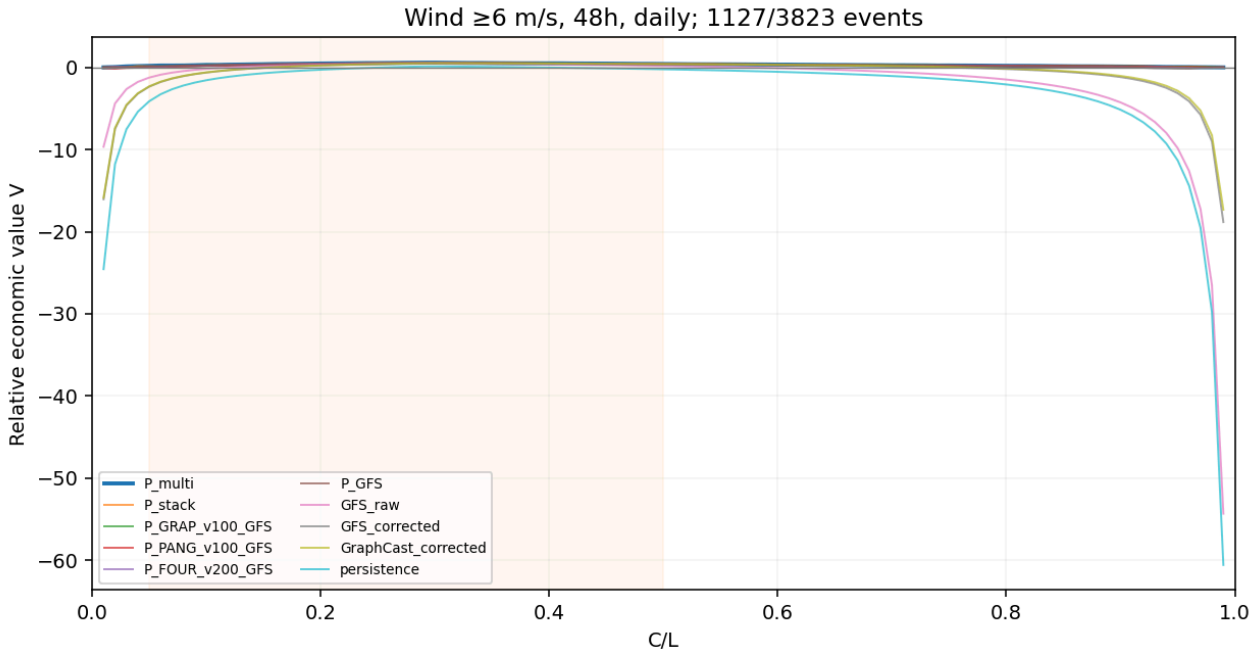
- CPA CIG/TCIG 2507 (2025) guidance on wind speeds for crane lifting, original PDF archived.
- Browell et al. 2016 ESREL: cost/loss + logistic probabilistic decision-making (offshore wind O&M), methodological precedent.
- Relative economic value V: Zhu et al. 2002 BAMS (journal p. 75, Eqs. 5–7); Richardson 2000 QJRMS (author preprint PDF p. 4 Eq. 3, journal page unverified); Murphy 1977 MWR (journal p. 808 Eq. 11). Card index.

Liang Hu Notes

Key figures

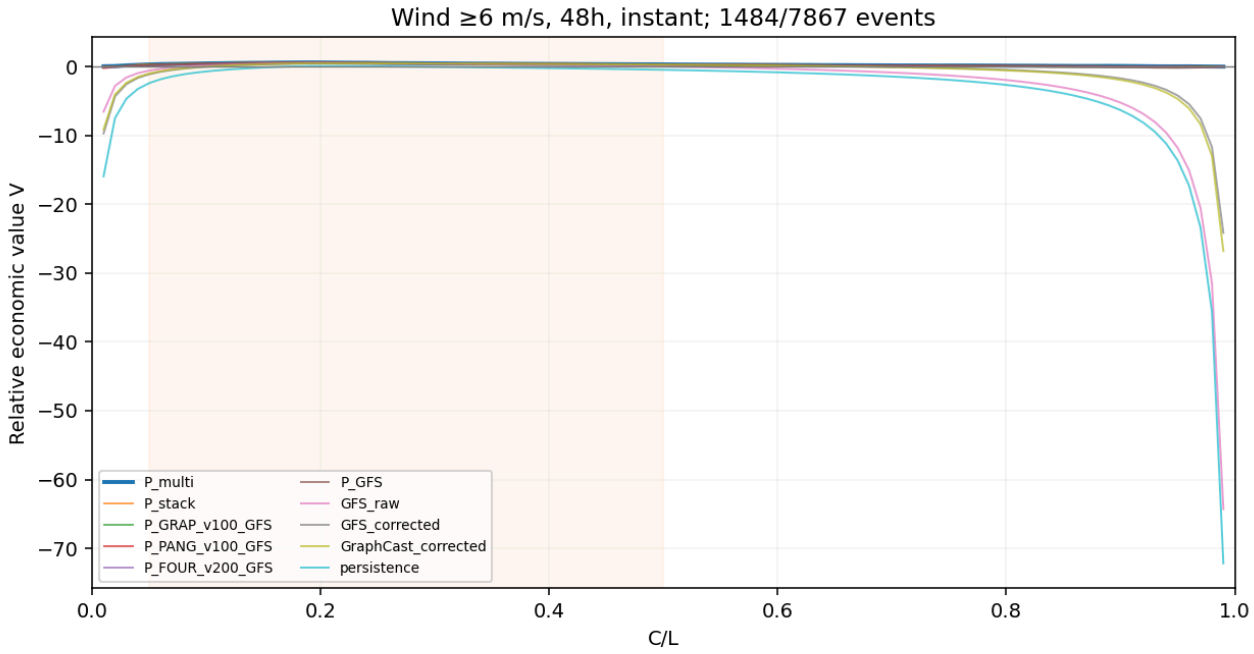


value curves wind 24h instant



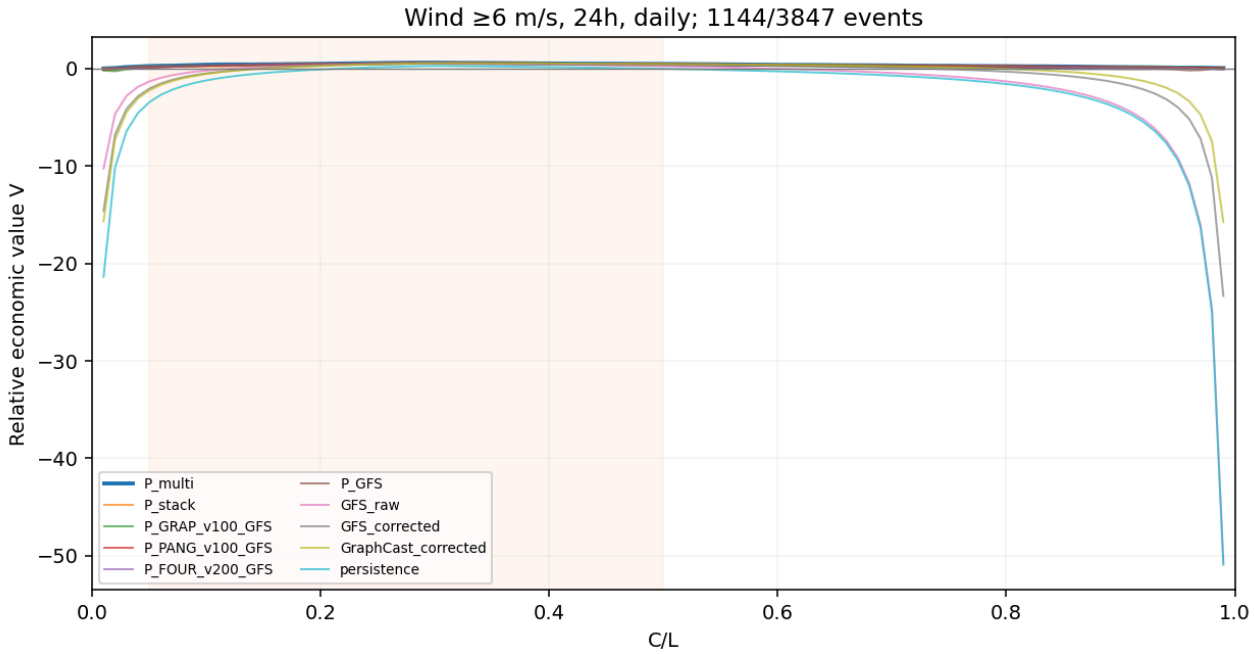
Data: ASOS/IEM; GraphCast/Pangu/FourCastNet v2 via AWS AIWP; NOAA GFS AWS. 2022 calibration, 2023 test.

value curves wind6 48h daily



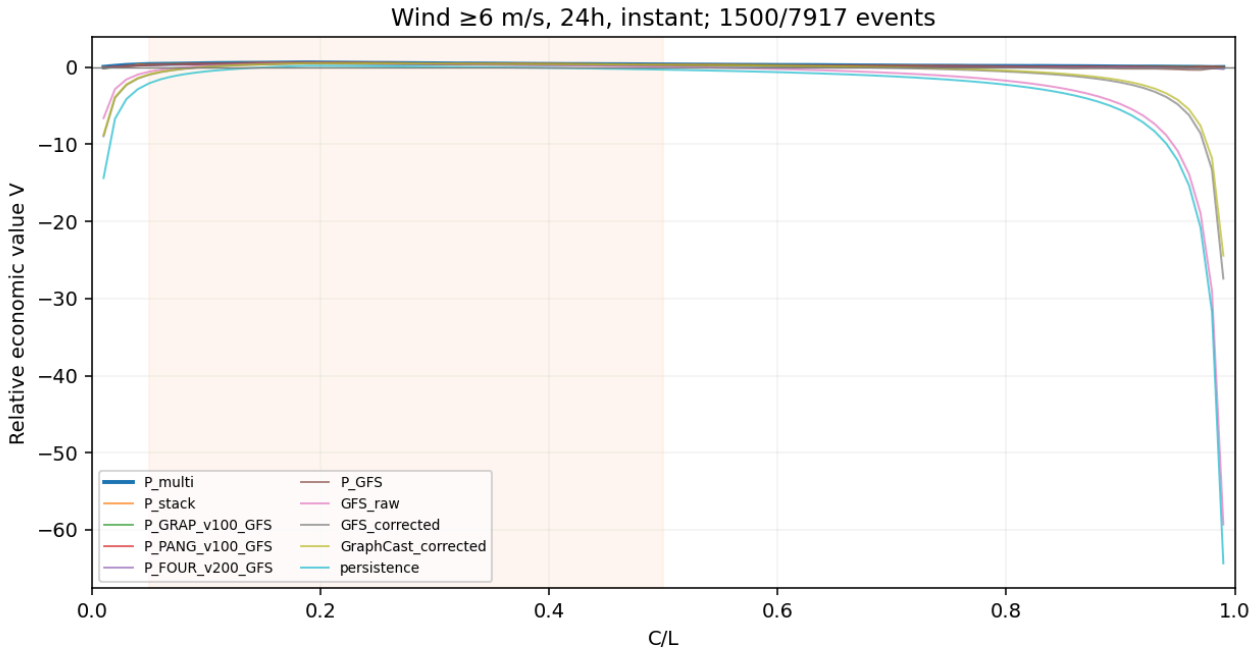
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Data: ASOS/IEM; GraphCast/Pangu/FourCastNet v2 via AWS AIWP; NOAA GFS AWS. 2022 calibration, 2023 test.

value curves wind6 24h daily



Data: ASOS/IEM; GraphCast/Pangu/FourCastNet v2 via AWS AIWP; NOAA GFS AWS. 2022 calibration, 2023 test.

value curves wind6 24h instant